

Advances in Post-selection Inference with E-values

Thesis Defense

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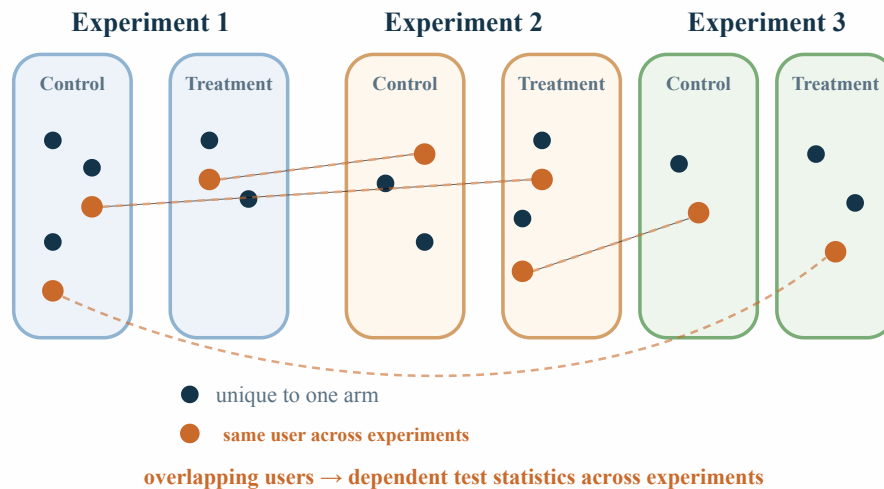
April 13, 2026



Multiple Testing Under Dependence

Scientists often test large numbers of hypotheses simultaneously and want to control false positives across their discoveries. But when experimental units are shared, the resulting test statistics are dependent.

A/B testing platforms

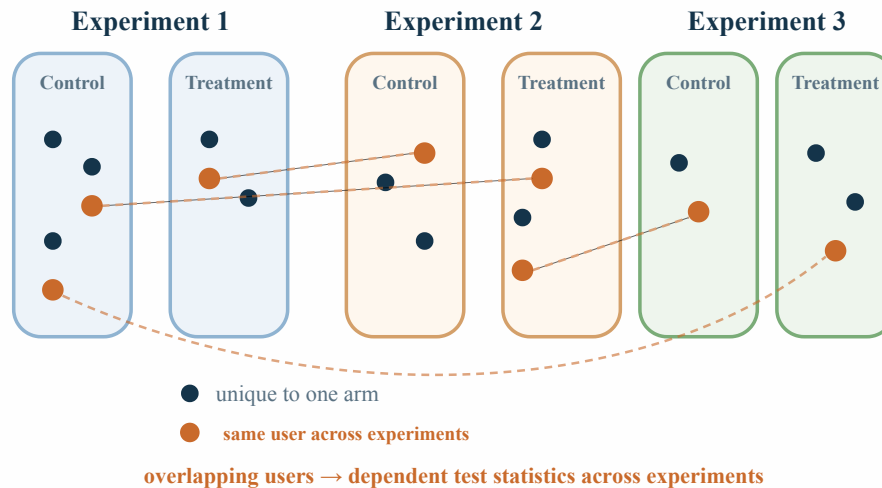


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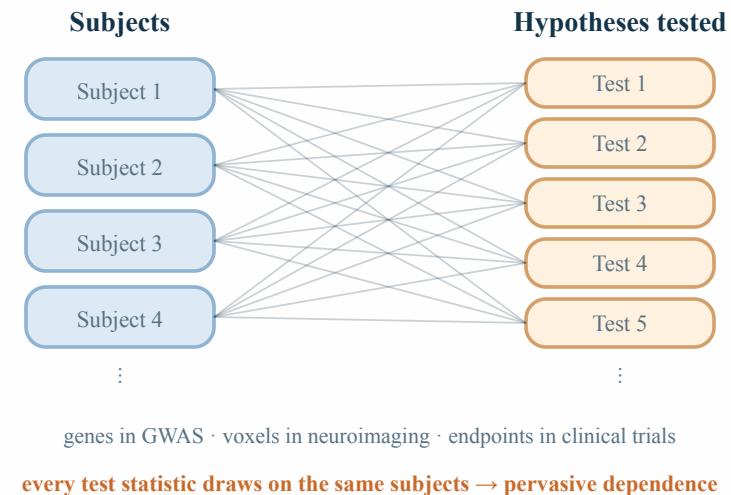
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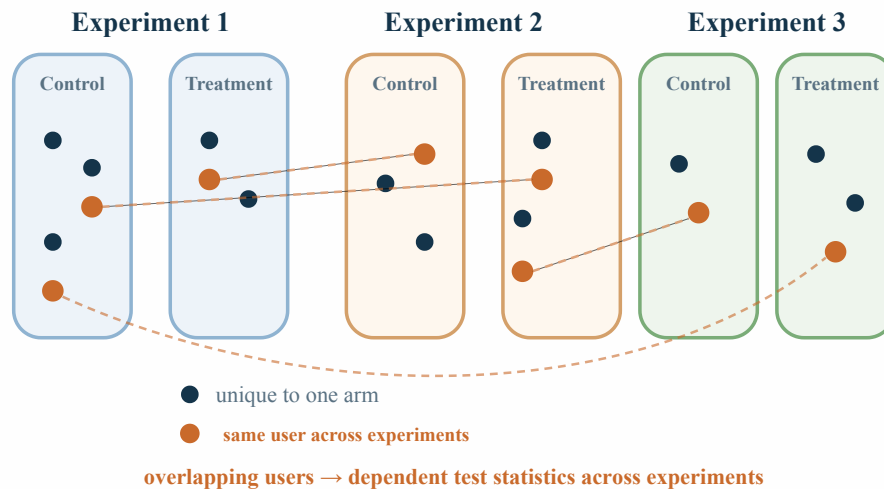


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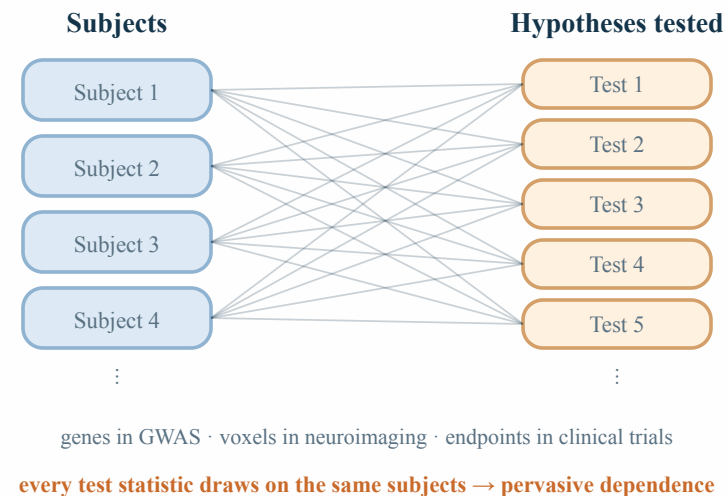
Scientists often test large numbers of hypotheses simultaneously and want to control false positives across their discoveries. But when experimental units are shared, the resulting test statistics are dependent.

A/B testing platforms



The same users appear in treatment and control arms of different experiments.

Genomics & neuroimaging



Every hypothesis is tested on the same subjects.

This thesis develops methodology to create valid and powerful multiple testing procedures under dependence.

Talk Focus

Part I — Ch. 2: e-Closure

- A generalized closure principle for expectation-based error control.
- Uniform improvements of existing multiple testing procedures with FDR control.
- A polynomial-time shortcut for computing the resulting procedures.

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Part III — Ch. 9: Donation

- Online e-Closure for online multiple testing.
- Online compound e-values and donation mechanism for improving procedures.
- Computationally efficient online multiple testing procedures that strictly improve over the status quo.

Background

Multiple Testing With Error Control

Definition: Null Sets, Intersection Nulls, and Rejection Families

$\mathcal{M}$ (universe of distributions) (i.e., distribution $\mathbf{P} \in \mathcal{M}$).

$H_1, \dots, H_m \subseteq \mathcal{M}$ (null hypotheses)

$N_{\mathbf{P}} = \{i \in [m] = \{1, \dots, m\} : \mathbf{P} \in H_i\}$ (true-null set at $\mathbf{P}$)

$H_S = \bigcap_{i \in S} H_i$ for $S \subseteq [m]$ (intersection hypothesis on S)

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Definition: Loss Function and Classical Error Rate

$\mathbf{f} : 2^{[m]} \times 2^{[m]} \rightarrow [0, \infty)$ (non-negative loss function)

$\text{ER}_{\mathbf{f}}(R) = \mathbb{E}_{\mathbf{P}}[\mathbf{f}_{N_{\mathbf{P}}}(R)]$ (classical error rate)

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Examples of General Risk Metrics

- family-wise error rate (FWER):

$$f_N(R) = 1\{|N \cap R| > 0\} \text{ and}$$

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- false discovery rate (FDR):**

$$f_N(R) = \text{FDP}_N(R) = \frac{|N \cap R|}{|R| \vee 1} \text{ and}$$

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- per-family error rate (PFER): $f_N(R) = |N \cap R|$

Prior Art: Benjamini-Hochberg and Benjamini-Yekutieli

Definition: p-value

For a null hypothesis H , a p-value P satisfies

$$\mathbb{P}(P \leq s) \leq s \text{ for all } s \in [0, 1] \text{ when } H \text{ is true.}$$

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With ordered p-values $P_{(1)} \leq \dots \leq P_{(m)}$, BH sets

$$\hat{k}_{\text{BH}} = \max\{k \in [m] : P_{(k)} \leq \alpha k/m\},$$

and rejects

$$R_{\alpha}^{\text{BH}} = \{i \in [m] : P_i \leq P_{(\hat{k}_{\text{BH}})}\}.$$

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Theorem (Benjamini–Hochberg 1995; Benjamini–Yekutieli 2001)

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Under arbitrary dependence of p-values $P_1, \dots, P_m$, $\text{FDR}(R_{\alpha}^{\text{BY}}) \leq \alpha.$

E-values, Self-consistency, and eBH

Definition: e-value

An e-value E satisfies $\mathbb{E}_{\mathbb{P}}[E] \leq 1$ under H . E^{-1} is a p-value by Markov's inequality: $\mathbb{P}(E \geq \alpha^{-1}) \leq \alpha$.

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The eBH procedure chooses the largest self-consistent rejection set:

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Reject the top $\hat{k}^{\text{eBH}}$ e-values:

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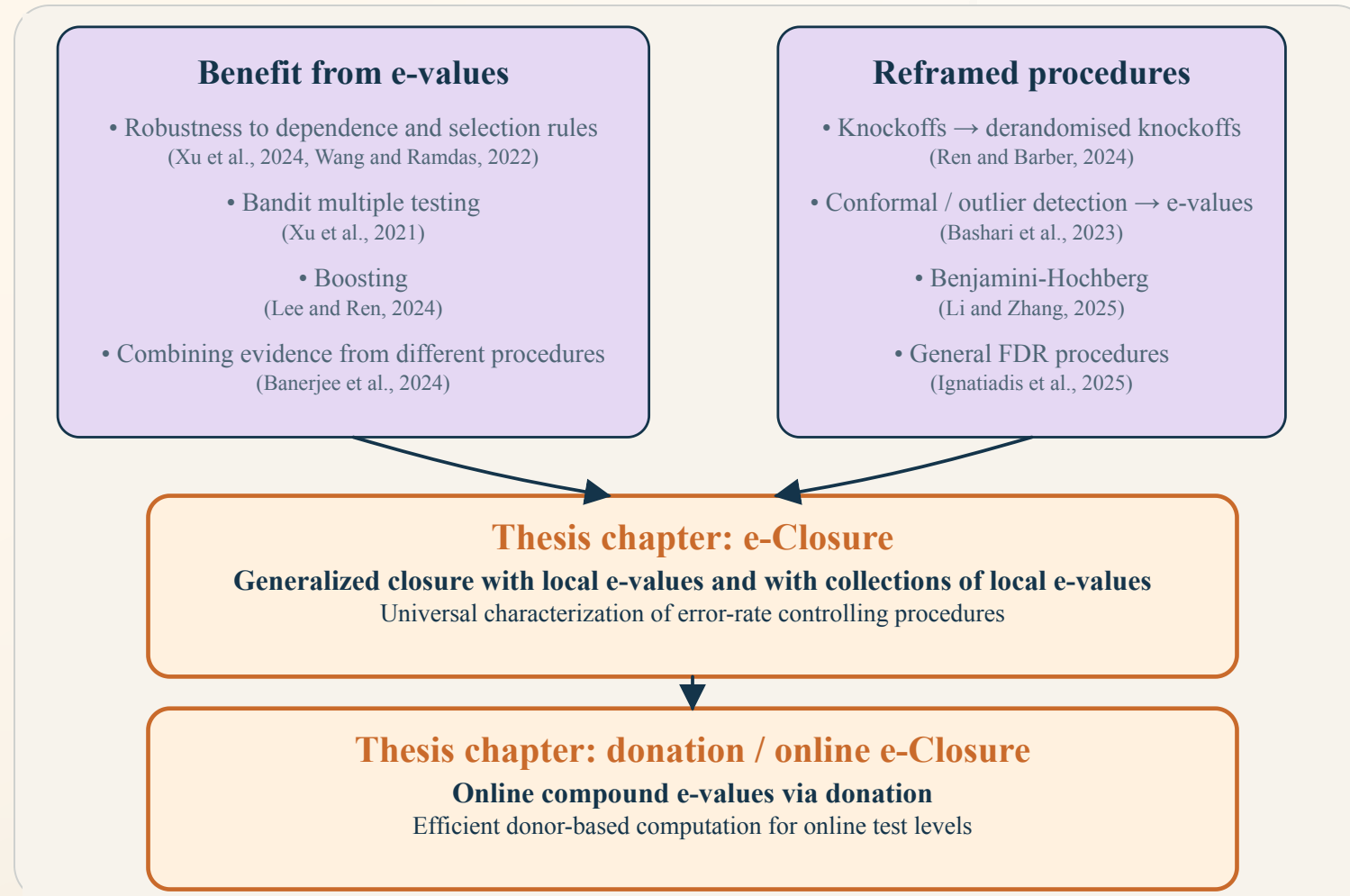
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Theorem (Wang and Ramdas, 2020)

Any level- α self-consistent procedure controls FDR under arbitrary dependence.

In particular, eBH inherits this guarantee because it is self-consistent.

Map of Related Work



PART I — BASED ON

Bringing Closure to False Discovery Rate Control: A General Principle for Multiple Testing

with Aldo Solari, Lasse Fischer, Rianne de Heide, Aaditya Ramdas, and Jelle Goeman

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Multiple Testing With General Risk Metrics

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Given loss f and valid family $\mathcal{R} \subseteq 2^{[m]}$, $\text{ER}_f(\mathcal{R}) = \mathbb{E}_{\mathbf{P}}[\max_{R \in \mathcal{R}} f_{N_{\mathbf{P}}}(R)]$ (simultaneous error rate)

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$$\mathbb{E}_{\mathbf{P}} \left[\max_{R \in \mathcal{R}(X)} f_{N_{\mathbf{P}}}(R) \right] = \sup_{\hat{R}: \hat{R}(x) \in \mathcal{R}(x) \text{ for all } x \in \mathcal{X}} \mathbb{E}_{\mathbf{P}} \left[f_{N_{\mathbf{P}}}(\hat{R}(X)) \right].$$

Prior Art: The Classical Closure Principle For FWER

Definition: Level- α Local Intersection Test

For an intersection null H_S , a local intersection test is a binary decision rule

$$\varphi_S \in \{0, 1\}.$$

It is a level- α local intersection test if

$$\mathbb{P}_{\mathbf{P}}(\varphi_S = 1) \leq \alpha \quad \text{for every } \mathbf{P} \in H_S.$$

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The Classical Closure Construction

For the family-wise error rate (FWER), the loss is

$$f_N(R) = 1\{|N \cap R| > 0\}.$$

Classical closure builds a level- α local intersection test φ_S for every H_S , then declares

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Theorem (Marcus, Peritz, and Gabriel, 1976; Sonnemann, 1982)

R controls FWER at level $\alpha \iff R = R(\varphi)$ for some family of level- α local tests.

Bonferroni To Holm Via Closure

Bonferroni Local Tests

Take Bonferroni local tests

$$\varphi_S^B = 1 \left\{ \min_{i \in S} P_i \leq \frac{\alpha}{|S|} \right\}.$$

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Holm as the Closure of Bonferroni

Then the direct Bonferroni rule is

$$R^B = \left\{ i : P_i \leq \frac{\alpha}{m} \right\},$$

while the closure of those same local tests is Holm's procedure:

$$R^{\text{Holm}} = R(\varphi^B).$$

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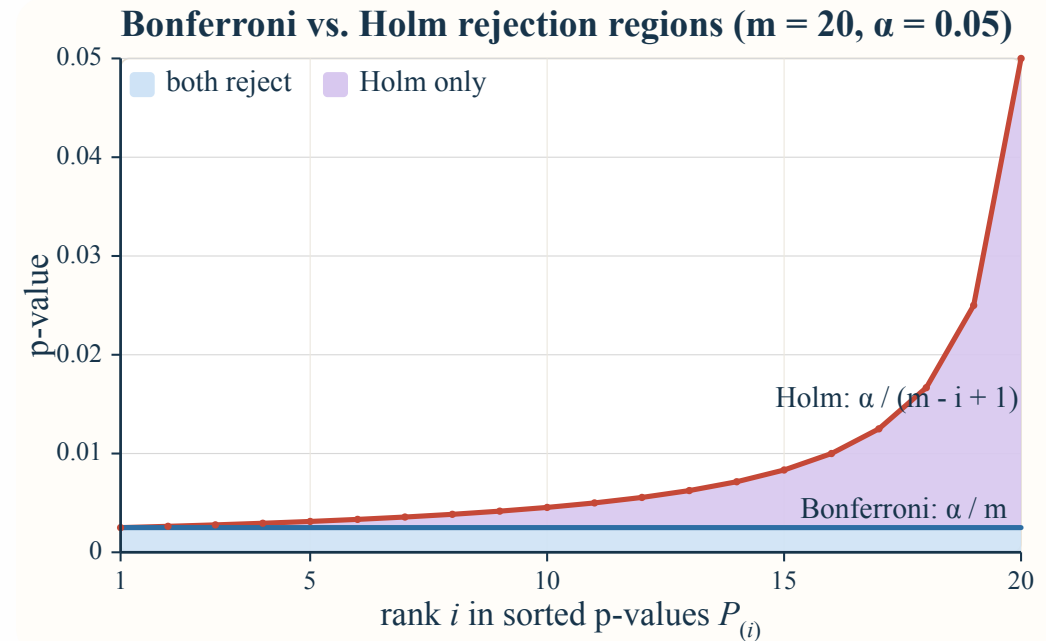
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The purple band is the gain from closure: ranks whose sorted p-values fail Bonferroni but satisfy Holm.

E-collection and the resulting simultaneous discovery set

Definition: e-collection

A subset-indexed family $\mathbf{E} = (E_S)_{S \subseteq [m]}$ is an e-collection if $\mathbb{E}_{\mathbf{P}}[E_{N_{\mathbf{P}}}] \leq 1$ for all $\mathbf{P} \in \mathcal{M}$, where $\mathcal{M}$ is the universe of distributions.

An e-collection measures evidence against each possible null configuration S simultaneously.

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An e-collection measures evidence against each possible null configuration S simultaneously.

The e-Closure simultaneous discovery set from an e-collection

Given an e-collection and a nonnegative loss $f : 2^{[m]} \times 2^{[m]} \rightarrow [0, \infty)$, the e-Closure simultaneous discovery set is

$$\mathcal{R}_{\alpha}^{\text{ER}_f}(\mathbf{E}) = \left\{ R \subseteq [m] : E_S \geq \frac{f_S(R)}{\alpha} \text{ for all } S \subseteq [m] \right\}.$$

Note that $R \in \mathcal{R}_{\alpha}^{\text{ER}_f}(\mathbf{E}) \implies f_{N_{\mathbf{P}}}(R) \leq \alpha E_{N_{\mathbf{P}}}.$

The Generalized E-Closure Principle

Theorem (ours)

For any nonnegative loss $f : 2^{[m]} \times 2^{[m]} \rightarrow [0, \infty)$, simultaneous error rate control is equivalent to being dominated by the e-Closure simultaneous discovery set of a subset-indexed e-collection.

$$\mathcal{R} \text{ controls } \text{ER}_f \text{ at level } \alpha \iff \text{there exists } \mathbf{E} \text{ such that } \mathcal{R} \subseteq \mathcal{R}_\alpha^{\text{ER}_f}(\mathbf{E}).$$

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Proof

Forward direction

$$\begin{aligned} R &\in \mathcal{R}_\alpha^{\text{ER}_f}(\mathbf{E}), \quad S = N_P. \\ f_{N_P}(R) &\leq \alpha E_{N_P} \implies \mathbb{E}_P[f_{N_P}(R)] \leq \alpha. \end{aligned}$$

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The Generalized E-Closure Principle

Theorem (ours)

For any nonnegative loss $f : 2^{[m]} \times 2^{[m]} \rightarrow [0, \infty)$, simultaneous error rate control is equivalent to being dominated by the e-Closure simultaneous discovery set of a subset-indexed e-collection.

$$\mathcal{R} \text{ controls } \text{ER}_f \text{ at level } \alpha \iff \text{there exists } \mathbf{E} \text{ such that } \mathcal{R} \subseteq \mathcal{R}_\alpha^{\text{ER}_f}(\mathbf{E}).$$

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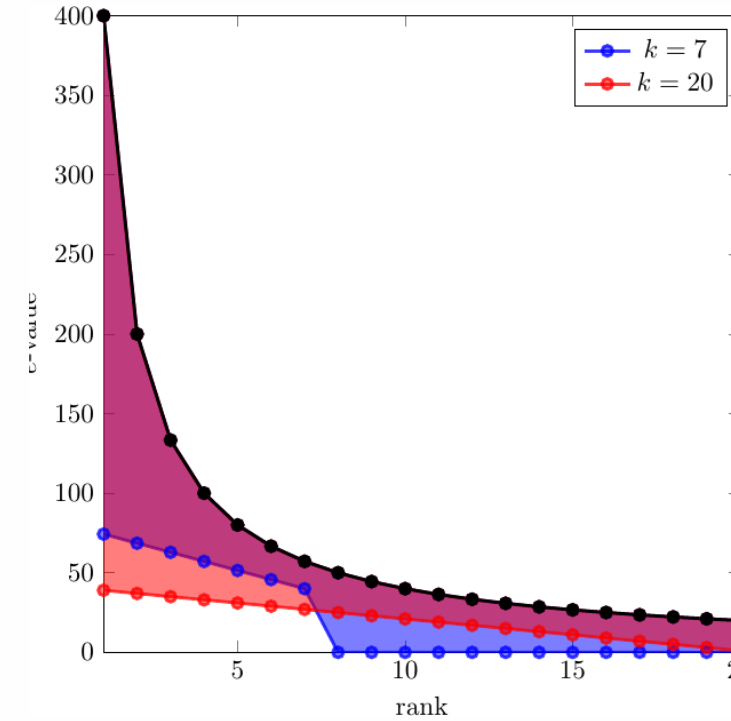
Hence every $R \in \mathcal{R}$ is feasible, so $\mathcal{R} \subseteq \mathcal{R}_\alpha^{\text{ER}_f}(\mathbf{E})$.

$\overline{\text{eBH}}$: Construction And Regions

The $\overline{\text{eBH}}$ Construction

Assume m arbitrarily dependent e-values $E_1, \dots, E_m$. Use the admissible symmetric local e-value

$$E_S = \frac{1}{|S|} \sum_{i \in S} E_i.$$



Improvement region for $\overline{\text{eBH}}$ over eBH.

$\overline{\text{eBH}}$: Construction And Regions

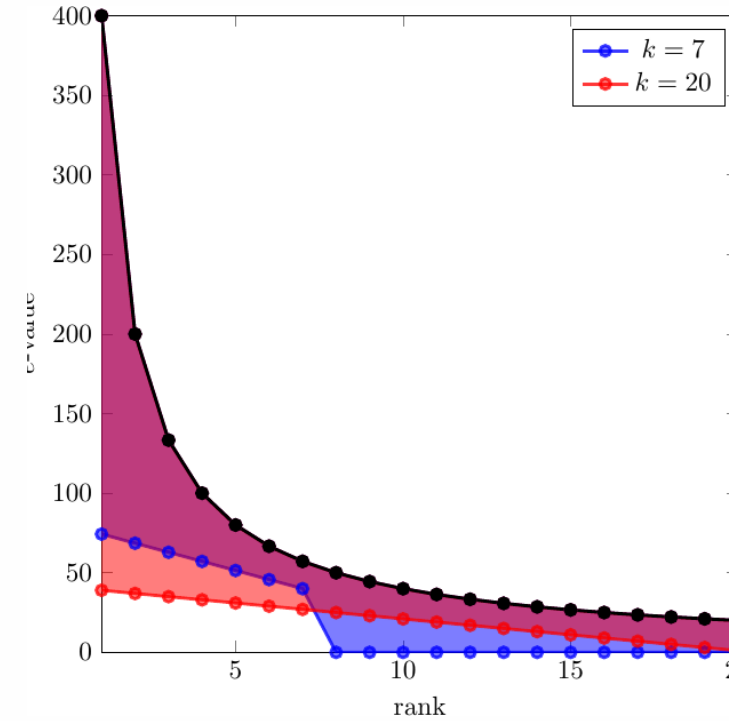
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$$\mathcal{R}_\alpha^{\overline{\text{eBH}}} = \left\{ R \subseteq [m] : \frac{1}{|S|} \sum_{i \in S} E_i \geq \frac{\text{FDP}_S(R)}{\alpha} \text{ for all } S \subseteq [m] \right\}.$$



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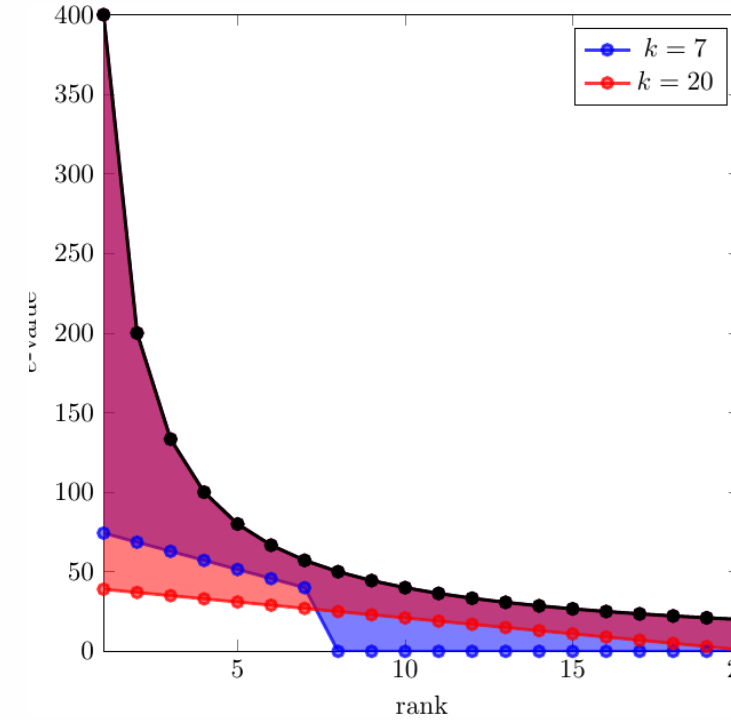
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Let $R_\alpha^{\overline{\text{eBH}}}$ be the set with the largest e-values that satisfies

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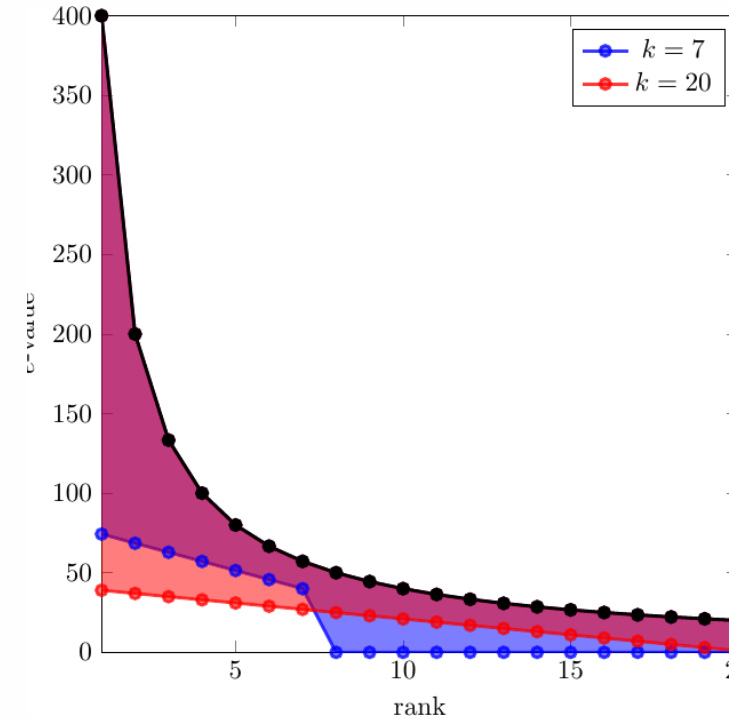
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Theorem (ours)

$\mathcal{R}_\alpha^{\overline{\text{eBH}}}$ controls FDR simultaneously at level α under arbitrary dependence among $E_1, \dots, E_m$.

Every self-consistent rejection set is mean-consistent, so

$R_\alpha^{\overline{\text{eBH}}} \in \mathcal{R}_\alpha^{\overline{\text{eBH}}}$ always. Therefore, $\overline{\text{eBH}}$ is a uniform improvement in both simultaneity and power over eBH.



Improvement region for $\overline{\text{eBH}}$ over eBH.

$\overline{\text{BY}}$: Construction And Regions

The $\overline{\text{BY}}$ Construction

Use the size-adaptive p-to-e calibrator

$$e_k(p) = \frac{k \mathbf{1}\{h_k p \leq \alpha\}}{\alpha(\lceil kh_k p / \alpha \rceil \vee 1)}.$$

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Define the local e-value and $\overline{\text{BY}}$ simultaneous discovery set

$$E_S = \frac{1}{|S|} \sum_{i \in S} e_{|S|}(P_i), \quad \mathcal{R}_\alpha^{\overline{\text{BY}}} = \mathcal{R}_\alpha^{\text{FDR}}(\mathbf{E}).$$

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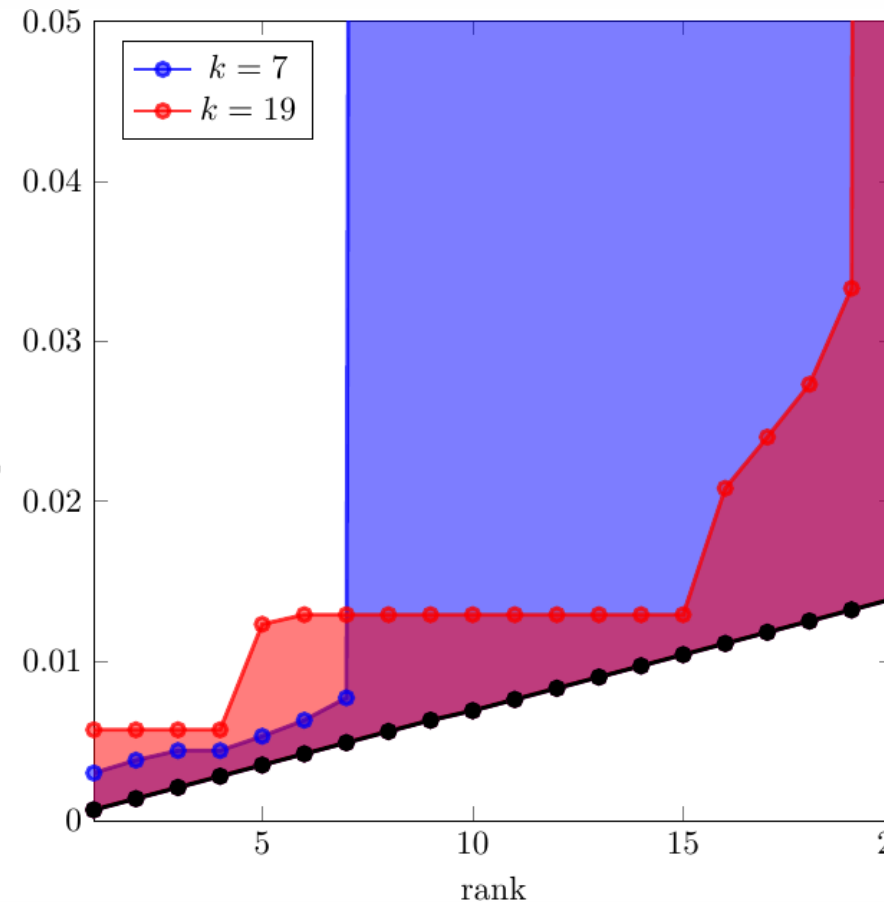
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Improvement region for $\overline{\text{BY}}$ over BY.

Computing E-Closure Procedures

Why Naive E-Closure Is Exponential

For a generic e-collection, checking $R \in \mathcal{R}_\alpha^{\text{FDR}}(\mathbf{E})$ is exponential in m , since every subset S may be relevant.

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For $\overline{\text{BY}}$ and $\overline{\text{Su}}$, structured monotonicity gives $O(m^3 \log m)$ shortcuts as well.

PART II — BASED ON

Improving Online FDR Procedures via Online Analogs of E-closure and Compound E-values

with Lasse Fischer and Aaditya Ramdas

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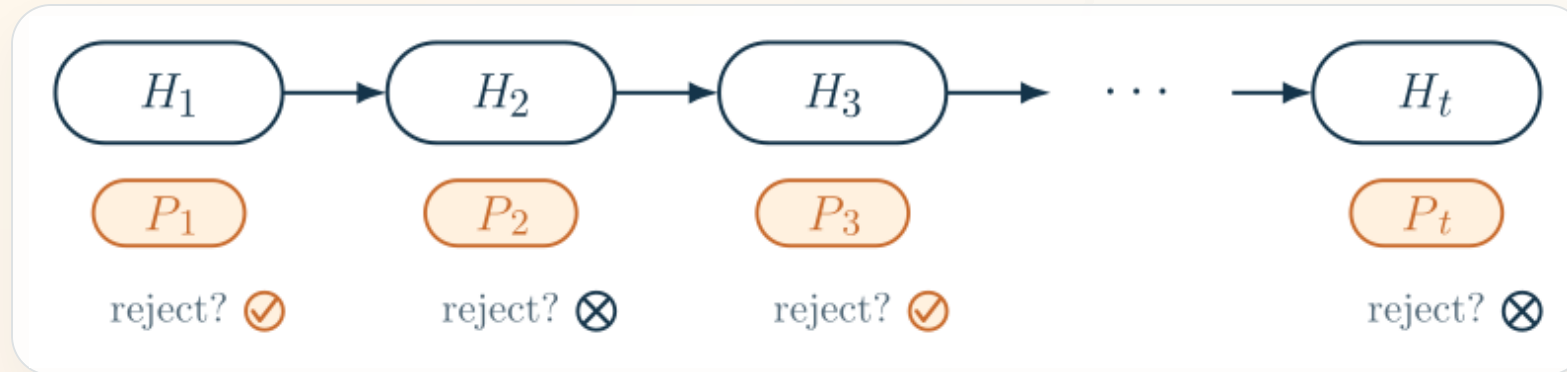
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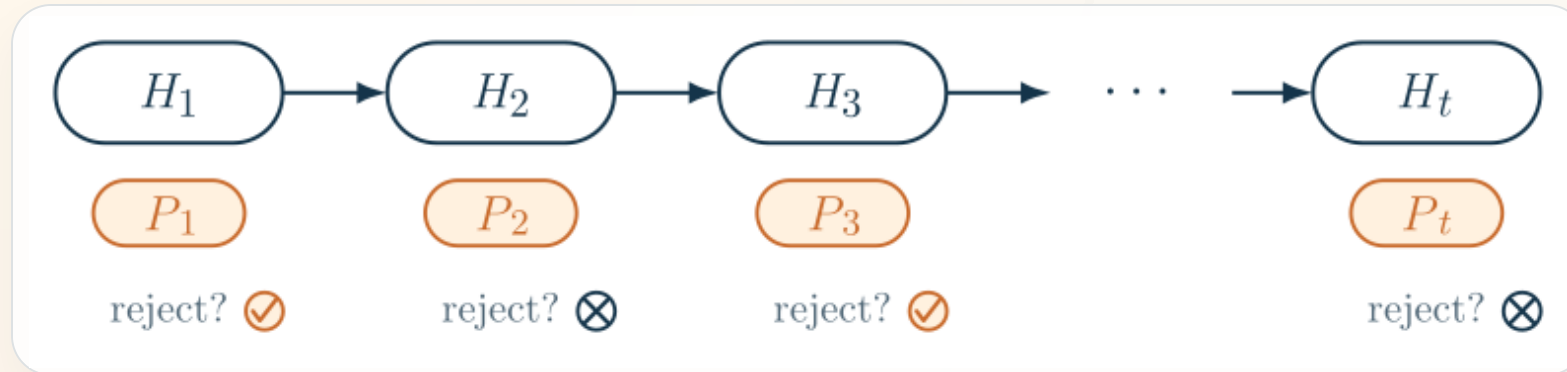
Online Testing Needs Stronger Error Metrics

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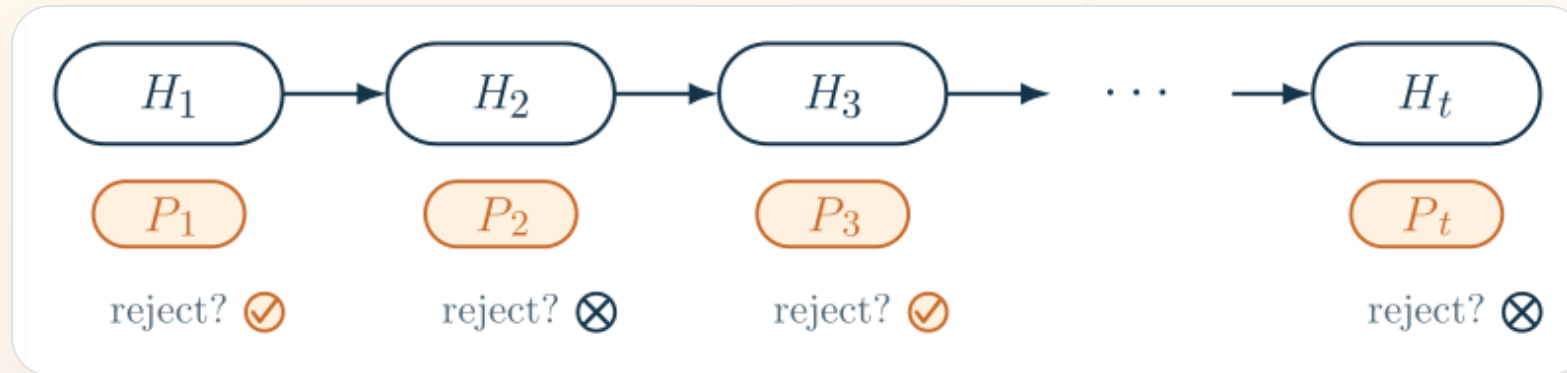
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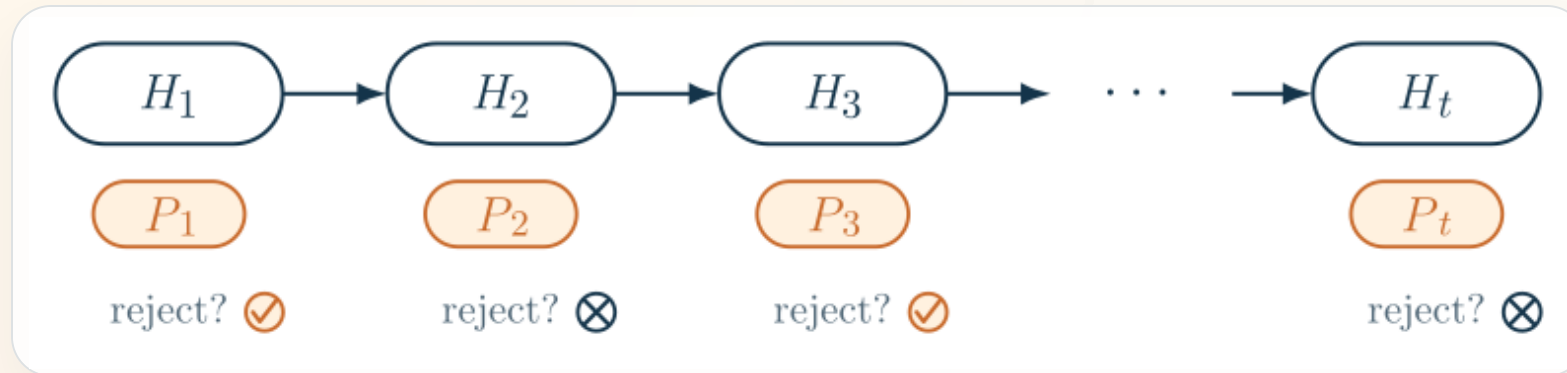


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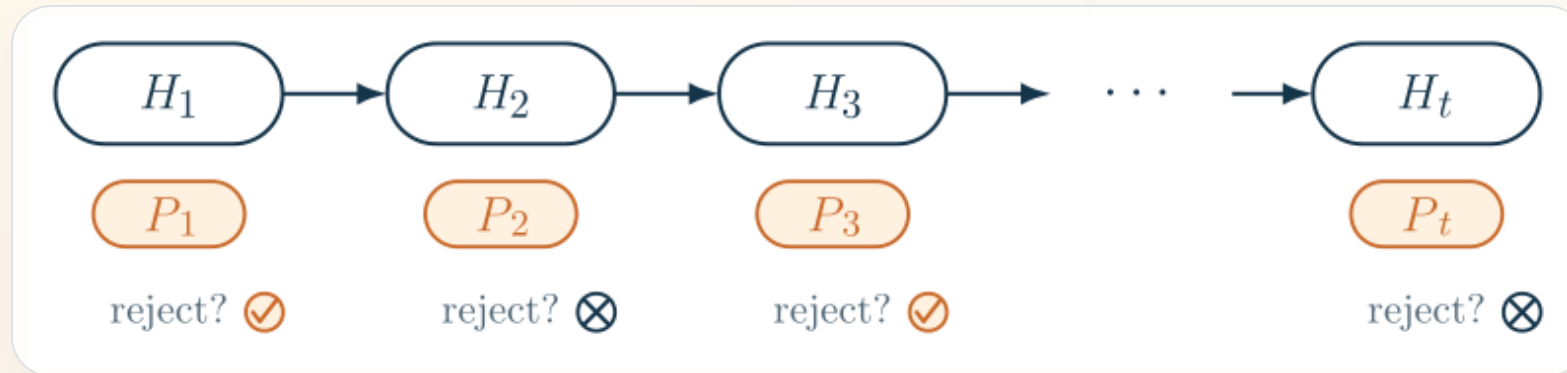
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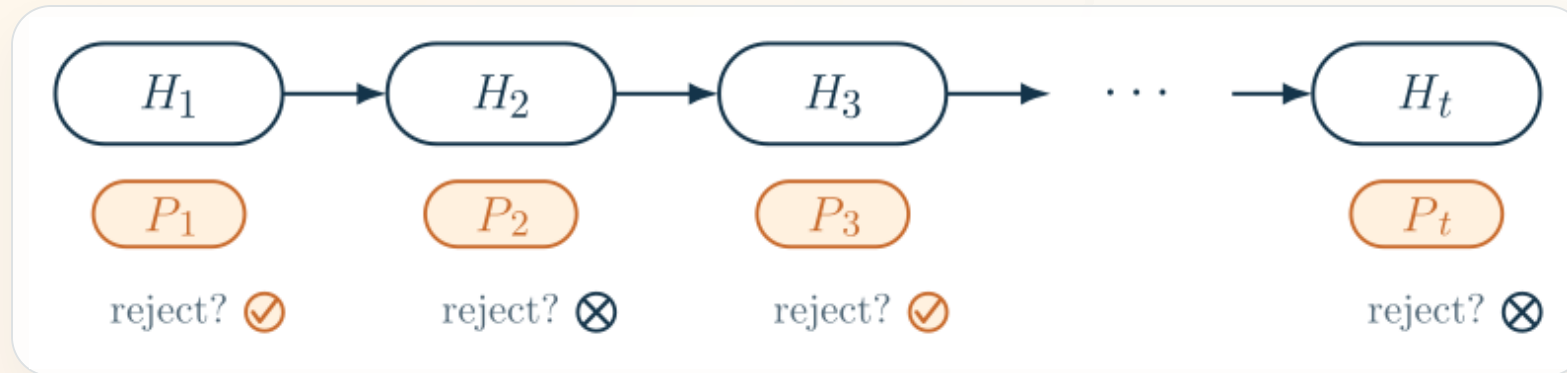
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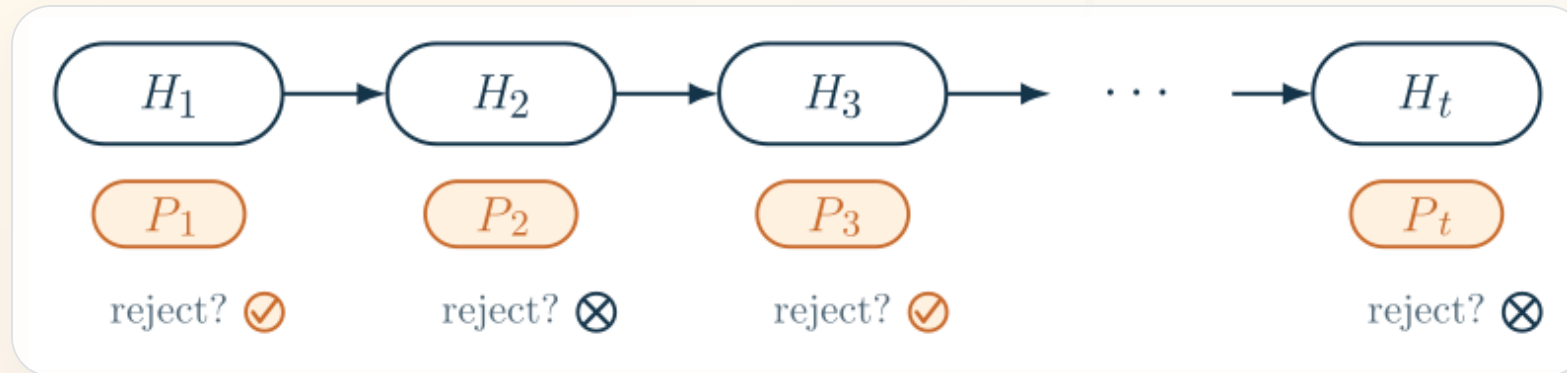
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SupFDR control implies StopFDR control, which implies onlineFDR control.

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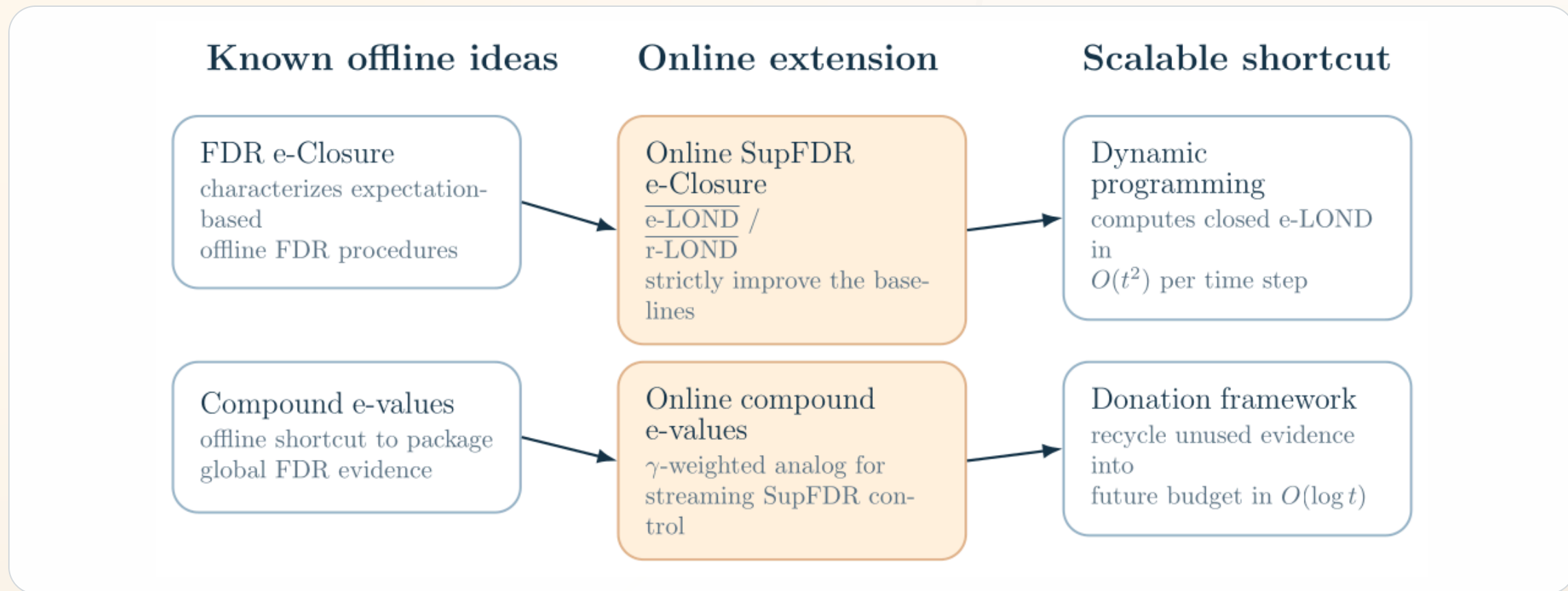
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We will focus on SupFDR in this work.

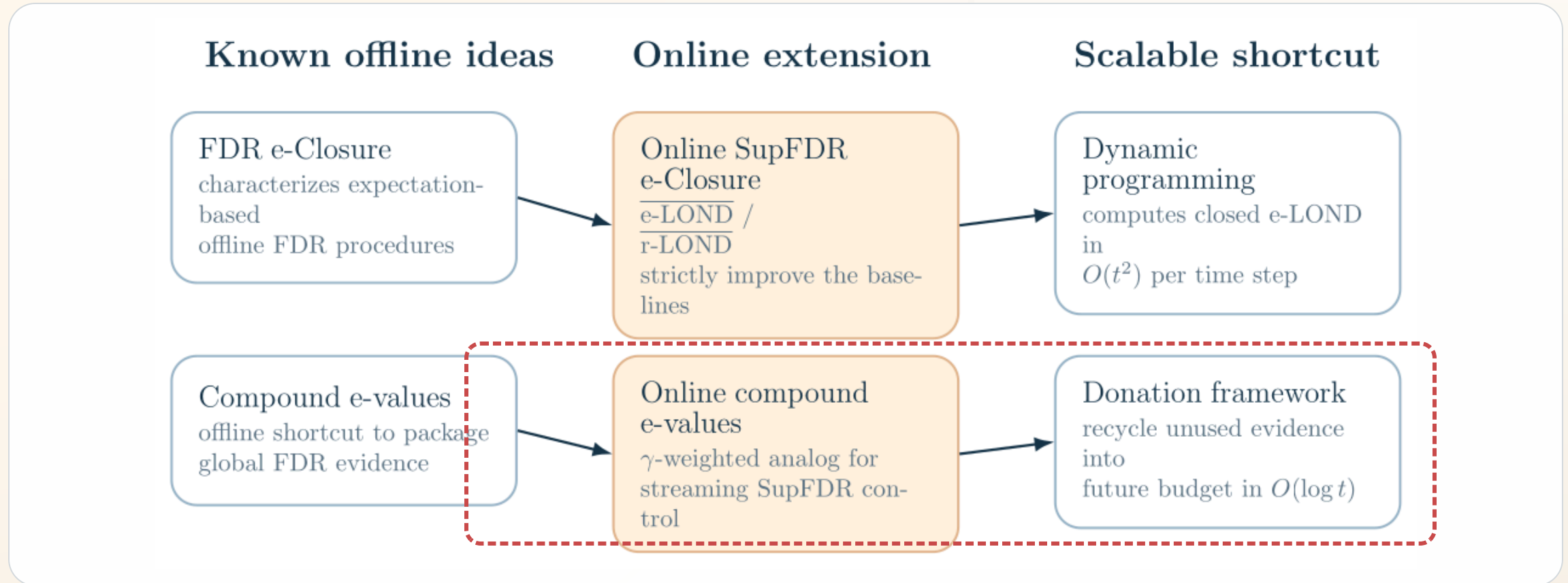


Donation Chapter Roadmap

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Prior Art: Online FDR Under Arbitrary Dependence

Definition: Weight Sequence

Fix $\gamma = (\gamma_t)_{t \geq 1}$ with

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e-LOND

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Theorems: LOND and r-LOND (Javanmard and Montanari 2015; Zrnic et al. 2021, Fischer et al. 2025)

- LOND: OnlineFDR control at level α under independence or PRDS.
- r-LOND: SupFDR control at level α under arbitrary dependence of p-values.

Online Compound E-values

Definition: Online Compound e-values

A stream $\tilde{E}_1, \tilde{E}_2, \dots$ of nonnegative random variables is a sequence of γ -online compound e-values if

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Theorem: Weighted Self-consistency (ours)

The weighted self-consistent family

$$\mathcal{C} = \left\{ R \subset \mathbb{N} : \tilde{E}_t \geq \frac{1}{\alpha \gamma_t |R|} \text{ for all } t \in R \right\}.$$

Applied to γ -online compound e-values, it satisfies

$$\mathbb{E} \left[\sup_{R \in \mathcal{C}} \text{FDP}(R) \right] \leq \alpha.$$

We construct such sequences via donation on the next slide.

Donation Gives γ -Online Compound E-values

Definition: γ -weighted donation

Let $E_1, E_2, \dots$ be e-values. A sequence $B_1, B_2, \dots$ is a valid γ -weighted donation if

$$B_t \geq -(E_t \wedge 1) \quad \text{and} \quad \sum_{i \leq t} \gamma_i B_i \leq 0 \quad \text{for all } t.$$

Define the donation e-values by

$$\tilde{E}_t := E_t + B_t.$$

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Proposition (ours)

If $(E_t)_{t \geq 1}$ are e-values and $(B_t)_{t \geq 1}$ is a valid γ -weighted donation, then $(\tilde{E}_t)_{t \geq 1}$ is a sequence of γ -online compound e-values:

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Proof

Write

$$\sum_{i \in N} \gamma_i \mathbb{E}[\tilde{E}_i] = \sum_{i \in N} \gamma_i \mathbb{E}[E_i] + \sum_{i \in N} \gamma_i \mathbb{E}[B_i].$$

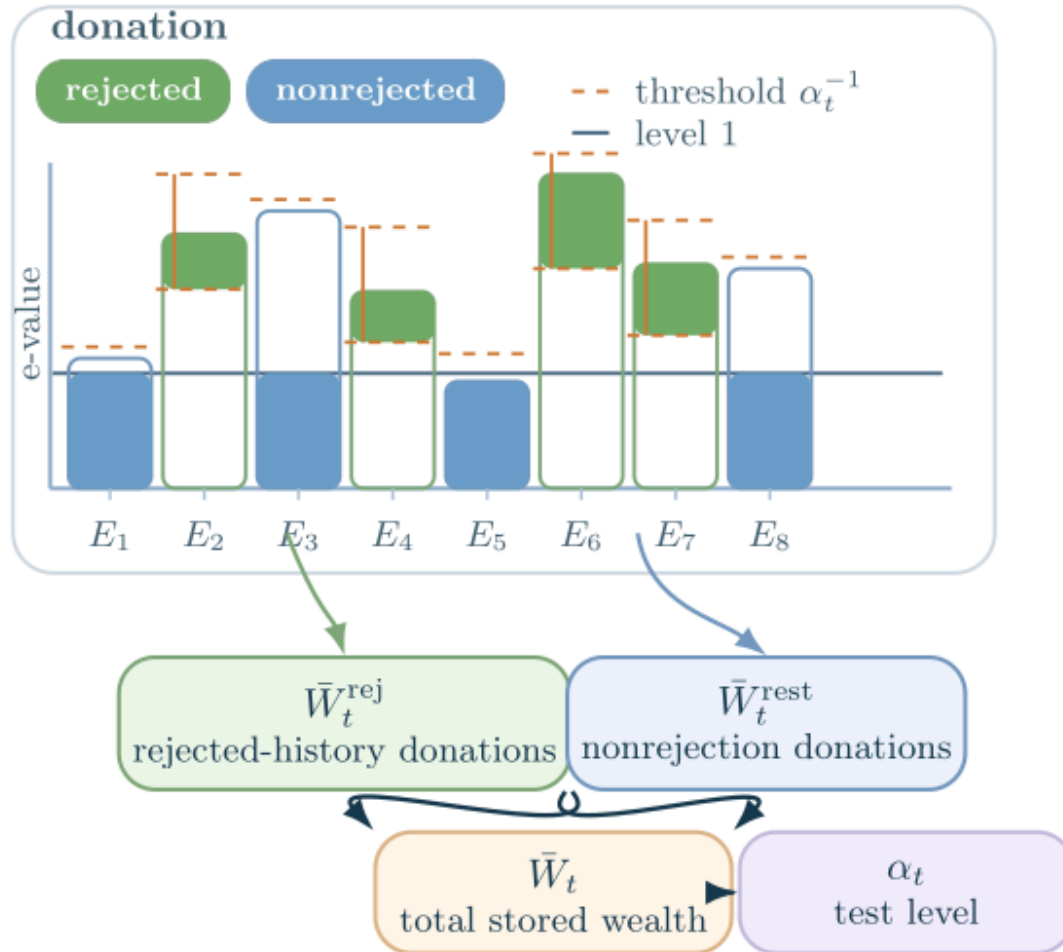
The e-value term is at most 1, since (E_t) is a sequence of e-values. The donation constraint gives

$$\sum_{i \in N} \gamma_i B_i \leq \sum_{i \in N \setminus N} \gamma_i.$$

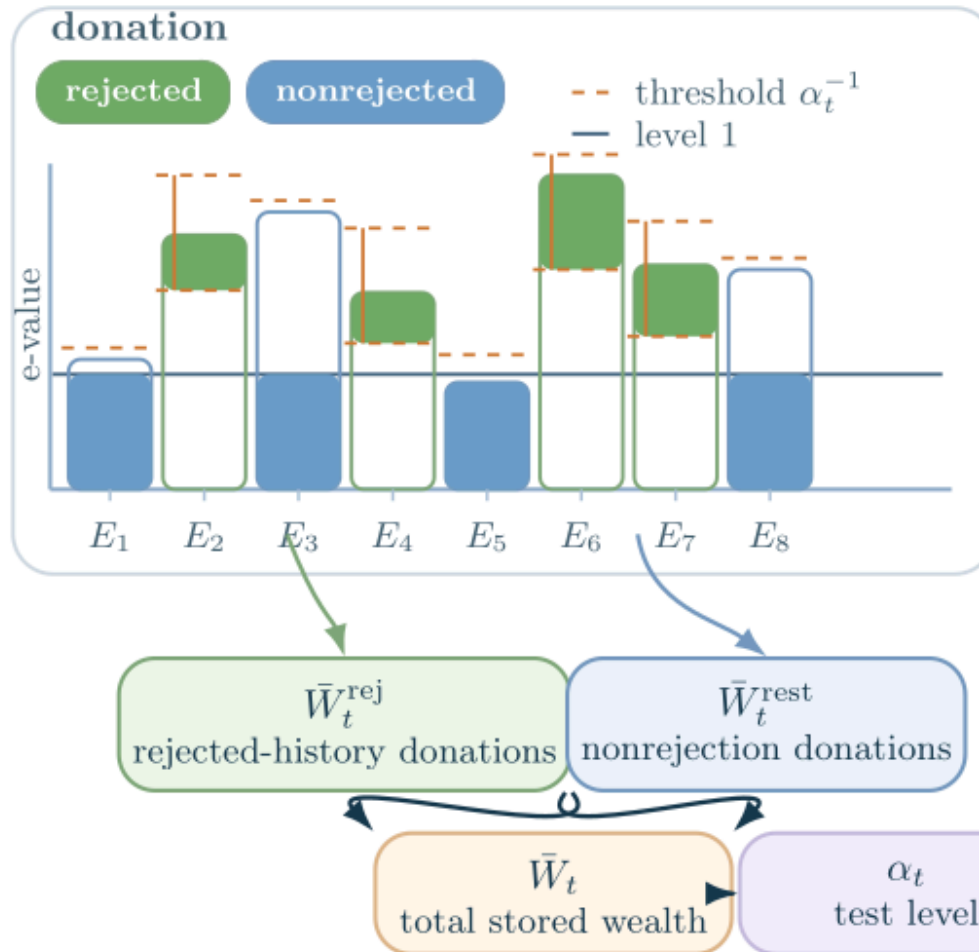
Hence

$$\sum_{i \in N} \gamma_i \mathbb{E}[\tilde{E}_i] \leq 1.$$

Donation E-values



Donation E-values



Donation wealth

Let $r_t := |R_{t-1}|$.

Rejection wealth (from rejections):

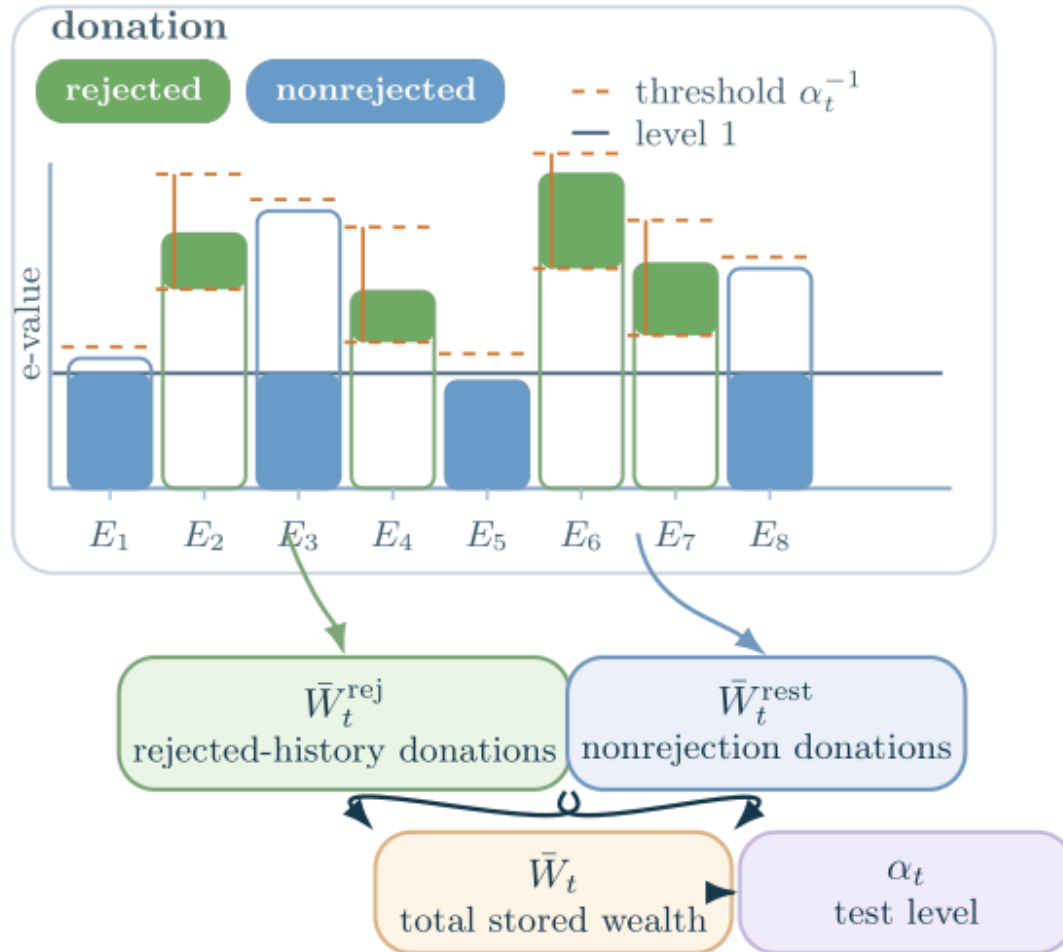
$$\bar{W}_t^{\text{rej}} = \sum_{i \in R_{t-1}} \gamma_i \left(\left(E_i - \frac{1}{\alpha \gamma_i (r_t + 1)} \right) \wedge 1 \right).$$

Remaining wealth (from nonrejections):

$$\bar{W}_t^{\text{rest}} = \sum_{i \notin R_{t-1}} \gamma_i (E_i \wedge 1).$$

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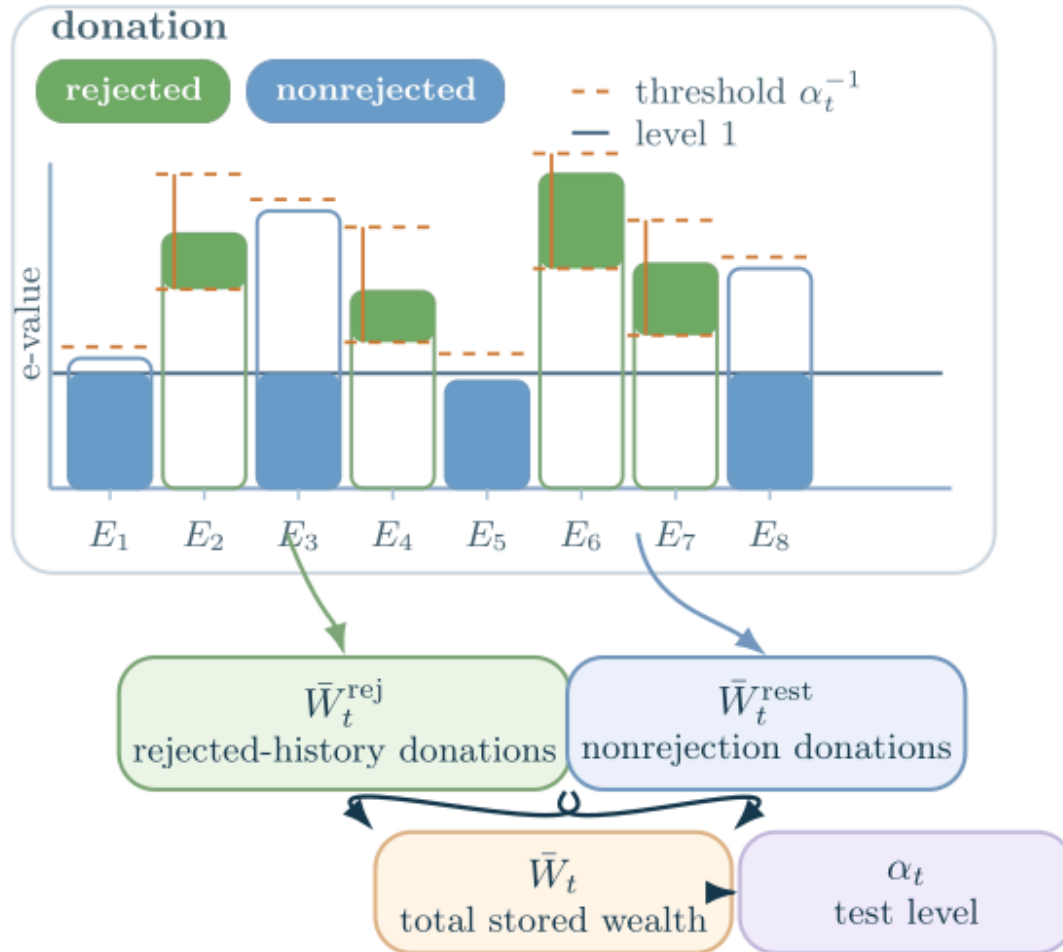
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$$\alpha_t^{\text{d-eLOND}} = \frac{\alpha \gamma_t (r_t + 1)}{1 - (\alpha (r_t + 1) \bar{W}_t \wedge 1)}.$$

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Theorem (ours)

SupFDR control at level α under arbitrary dependence.

Strictly improves over e-LOND.

Computational Efficiency of Donation

Donation e-LOND

Maintain a rejection tree keyed by $\gamma_i(E_i - 1)$; one split at $\hat{\alpha}_t - 1$ exposes cached left/right summaries for the next level.

$$\hat{\alpha}_t^{-1} = \frac{1}{\alpha(r_t + 1)}.$$

$$\gamma_i(E_i - 1) < \hat{\alpha}_t - 1 \implies \gamma_i(E_i - \hat{\alpha}_t), \quad \gamma_i(E_i - 1) \geq \hat{\alpha}_t - 1 \implies \gamma_i.$$

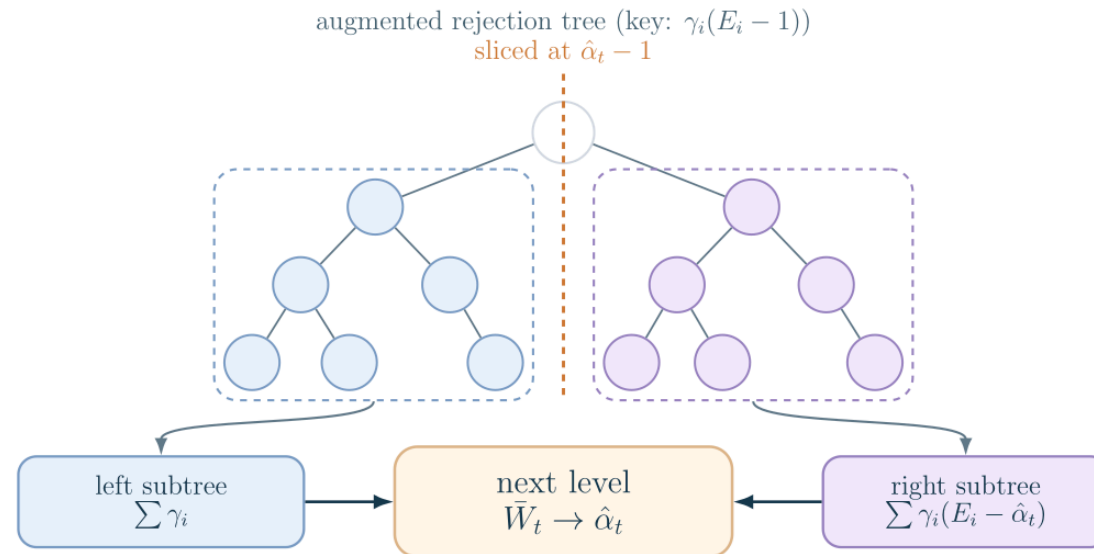
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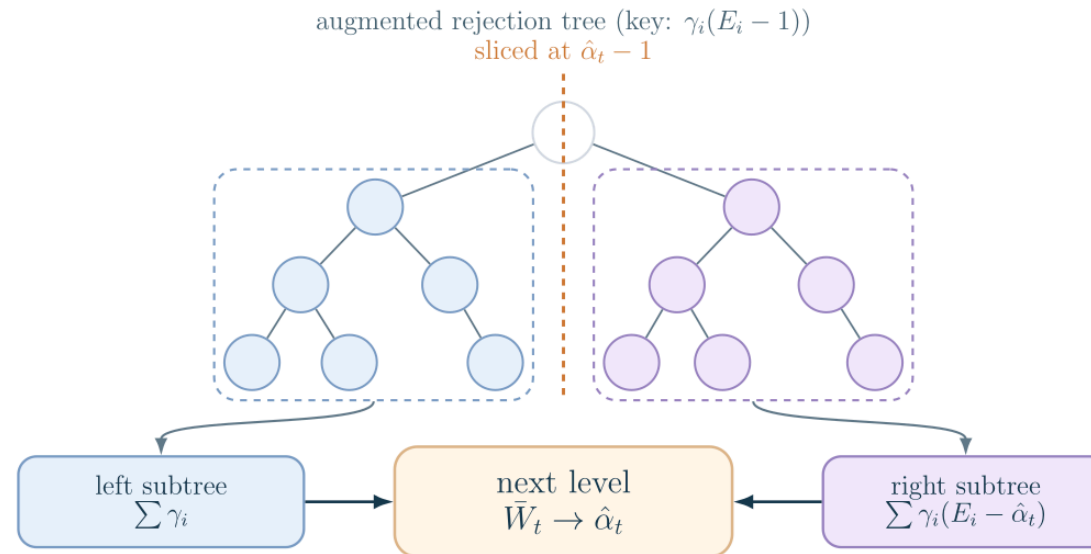
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Result: computing one donation e-LOND level uses one balanced-tree split plus cached subtree summaries, so the per-step cost is $\mathcal{O}(\log t)$.

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Main takeaways

- **e-Closure** generalizes classical closure to expectation-based multiple testing via e-values.

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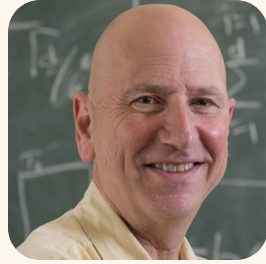
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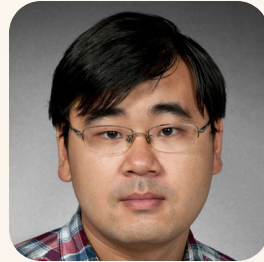
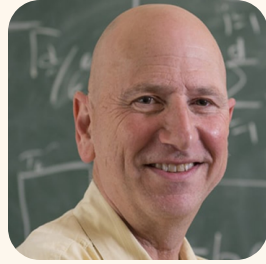
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- Can we derive a characterization of admissible procedures directly?
- How do we generalize multiple testing techniques to other decision-theoretic utilities (e.g., sample size investment)?

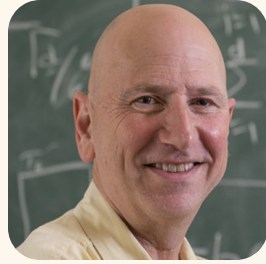
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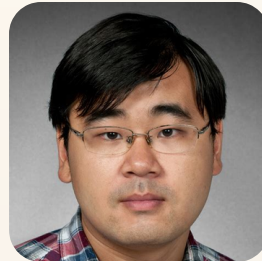
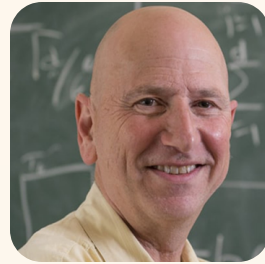
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Thanks to all my collaborators and people who have helped me throughout my PhD!

Appendix



Why $\overline{\text{eBH}}$ Contains eBH

Theorem (ours)

$\mathcal{R}_\alpha^{\overline{\text{eBH}}}$ controls FDR simultaneously at level α under arbitrary dependence among $E_1, \dots, E_m$.

Every self-consistent rejection set is mean-consistent, so $R_\alpha^{\text{eBH}} \in \mathcal{R}_\alpha^{\overline{\text{eBH}}}$ always. Therefore, $\overline{\text{eBH}}$ is a uniform improvement in both simultaneity and power over eBH .

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If R is self-consistent, then for any subset S ,

$$\frac{1}{|S|} \sum_{i \in S} E_i \geq \frac{|S \cap R|}{|S|} \min_{i \in R} E_i \geq \frac{\text{FDP}_S(R)}{\alpha}.$$

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- a hypothesis may be rejected even when $E_i < \alpha^{-1}$;
- the largest mean-consistent rejection set need not be unique.

Online SupFDR E-Closure: Setup

Definition: Increasing Online e-collections

An online e-collection $(E_S)_{S \in 2^{\mathbb{N}}}$ is **increasing** if

$$E_S \leq E_{S \cup S'} \quad \text{whenever } \min S' > \sup S.$$

Intuitively, appending later hypotheses to a null set can only increase the e-value. Each E_S is $\mathcal{F}_{\sup S}$ -measurable.

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The Candidate Family

Given such a collection, define the candidate rejection family at time t :

$$\mathcal{C}_t = \left\{ R \subseteq [t] : E_S \geq \frac{\text{FDP}_S(R)}{\alpha} \text{ for all } S \subseteq [t] \right\}.$$

This is the online analog of the e-Closure simultaneous discovery set $\mathcal{R}_\alpha^{\text{ER}_f}(\mathbf{E})$, adapted to the streaming setting.

Increasingness implies

$$\mathcal{C}_1 \subseteq \mathcal{C}_2 \subseteq \cdots,$$

which is what makes the stopping-time guarantee possible.

Online SupFDR E-Closure: Theorem

Theorem: Online SupFDR e-Closure Principle (ours)

Given an increasing e-collection $(E_S)_{S \in 2^{\mathbb{N}}}$ with E_S measurable w.r.t. $\mathcal{F}_{\text{sup}(S)}$, the associated e-closure collections $(\mathcal{C}_t)$ satisfy

$$\mathbb{E} \left[\sup_{t \in \mathbb{N}} \sup_{R \in \mathcal{C}_t} \text{FDP}_N(R) \right] \leq \alpha.$$

Consequently, any discovery sequence R with $R_t \in \mathcal{C}_t$ for all $t \in \mathbb{N}$ controls SupFDR at level α .

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What the Online Closure Theorem Gives

- this is the online analog of offline e-Closure;
- the target is SupFDR — the strongest stopping-time criterion;
- the construction separates statistical validity from later computational shortcuts.

e-LOND

The e-LOND Construction

For online e-Closure, weight each E_i by its order within S :

$$E_S = \sum_{i \in S} \gamma_{|S \cap [i]|} E_i.$$

$$\alpha_t^{\overline{\text{eLOND}}} = \min_{S \subseteq [t-1]} \frac{\alpha \gamma_{|S|+1} (|R_{t-1}| + 1)}{1 + |S \cap R_{t-1}| - \alpha E_S (|R_{t-1}| + 1)}.$$

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Strictly Improves over e-LOND

If γ_t is nonincreasing, then $1 + |S \cap R_{t-1}| - \alpha E_S (|R_{t-1}| + 1) \leq 1$ for every $S \subseteq [t-1]$, hence $\alpha_t^{\overline{\text{eLOND}}} \geq \alpha_t^{\text{eLOND}}$.

Computational Efficiency of Online e-Closure

e-LOND / online e-Closure

Refresh the reachable part of the $t \times k$ DP table, then minimize over the final row of $v(j, k)$.

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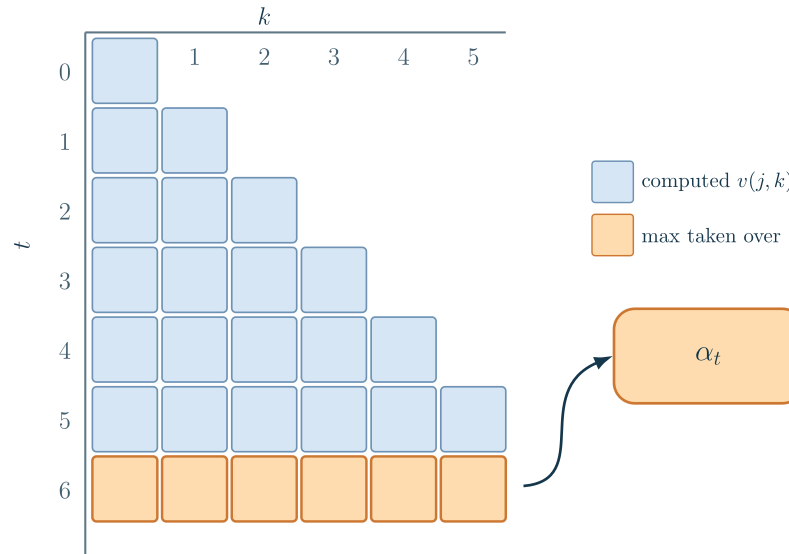
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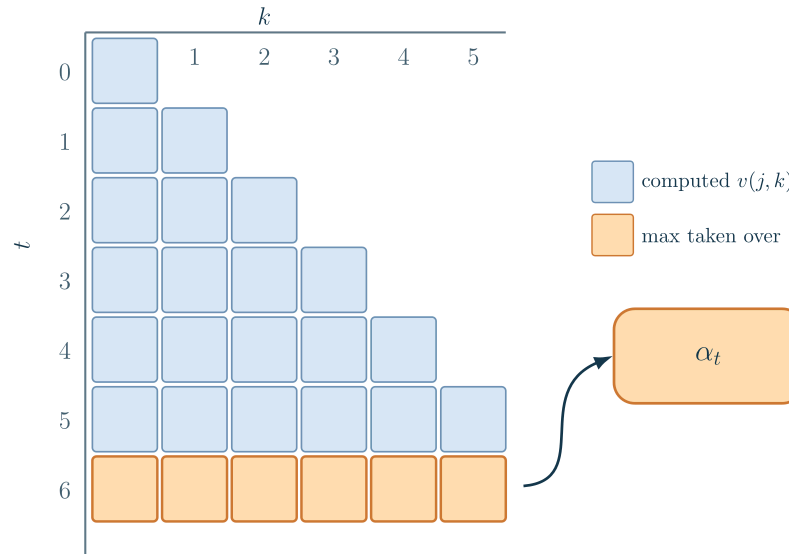
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Result: computing one e-LOND level needs a full $t \times t$ DP refresh and final-row minimization, so the per-step cost is $\mathcal{O}(t^2)$.

Extensions: e-Closure (in the paper)

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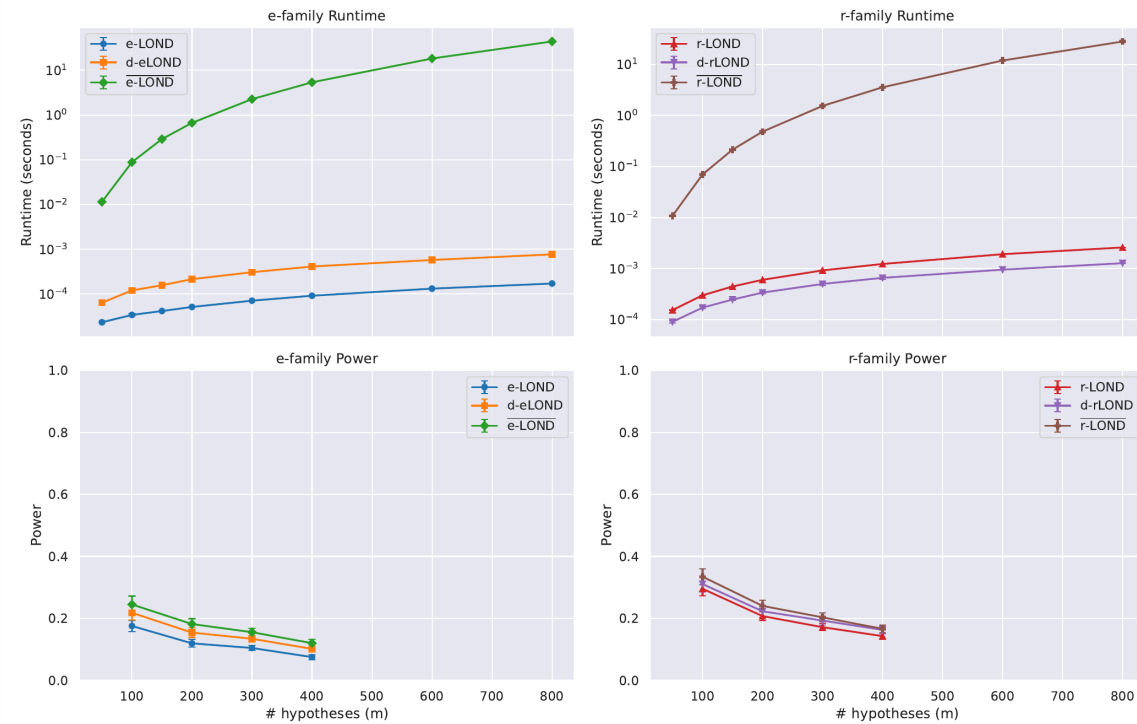
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- Admissible multiple testing with e-values

Simulations



Thesis Outline

Part I — Methodological advances via e-values

- Ch. 2: **e-Closure** — a general closure principle for FDR control
- Ch. 3: **Randomized e-values** — more powerful testing under dependence via randomization
- Ch. 4: **e-BY / FCR** — post-selection inference for e-value based confidence intervals

Part II — Adaptive data collection

- Ch. 5: **Active testing** — multiple testing with proxy p-values and e-values
- Ch. 6: **Bandit multiple testing** — a unified framework for bandit experiments

Part III — Online multiple testing

- Ch. 7: **e-LOND** — online multiple testing with e-values
- Ch. 8: **ARC** — an online generalization of the (e-)BH procedure
- Ch. 9: **Donation** — improving online FDR via online e-Closure and compound e-values